GTS123 Introduction to Computers and Programming

Lab 10: Problem 1

Write a Python program that accepts elements of a square matrix. Then, calculate the determinant, and the inverse of the input matrix (if possible).

The determinant should be displayed with two decimal places. If the determinant is zero, print `No inverse exists.`.

The output should look similar to the following.

```
Input size of matrix: 3
Input element at row 1 column 1: 1
Input element at row 1 column 2: 0
Input element at row 1 column 3: 0
Input element at row 2 column 1: 0
Input element at row 2 column 2: 1
Input element at row 2 column 3: 0
Input element at row 3 column 1: 0
Input element at row 3 column 2: 0
Input element at row 3 column 3: 1
Output:
Matrix =
[[1 0 0]
 [0 1 0]
 [0 0 1]]
Determinant = 1.0
Inverse matrix =
[[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]
```

```
Input size of matrix: 2
Input element at row 1 column 1: 4
Input element at row 1 column 2: 7
Input element at row 2 column 1: 2
Input element at row 2 column 2: 6
Output:
Matrix =
[[4. 7.]
 [2. 6.]]
Determinant = 10.0
Inverse matrix =
[[ 0.6 -0.7]
 [-0.2  0.4]]
```

```
Input a positive number: -5
Wrong Input
```

```
Input size of matrix: 2
Input element at row 1 column 1: 1
Input element at row 1 column 2: 2
Input element at row 2 column 1: 2
Input element at row 2 column 2: 4
Output:
Matrix =
[[1. 2.]
 [2. 4.]]
Determinant = 0.0
No inverse exists.
```

Note: To pass the test, your program must produce the exact output as shown above, including spaces and line breaks.